
LECTURE NOTE

Lecture 2: Vision Transformer

AI508 - Computervision

Simon Holm
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1 Tokenizations

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Figure 1: Tokenize by splitting the image up in patches. Note that these are commonly split into 16×16

2 Embedding

3 Swin transformer

- Window-based self-attention (W-MSA)

Uses patch merging: Same idea as in pooling, but instead merge patches, which attend to each other most so instead of attending globally, the image is split (by patch merging) into non-overlapping windows (e.g. 7×7 patches), and self-attention is computed only within each window. This makes more efficient complexity

- Shifted windows

Window attention alone means patches in different windows never interact. Swin alternates between two layouts across consecutive blocks: regular windows, then windows shifted. This shift creates cross-window connections in the next layer without the cost of global attention.

4 ViT code

Huggingface ViT [[link](#)]

Model github [[link](#)]